

An Investigation of the Selection of Texture Features for Crop Discrimination Using SAR Imagery

João Vianeí Soares,* Camilo Daleles Rennó,* Antonio Roberto Formaggio,*
Corina da Costa Freitas Yanasse,* and Alejandro Cesar Frery*

This article presents a methodology for selecting texture measures to maximize the discrimination of agricultural land use classes in SAR images. The images were acquired during the first flight of the Shuttle Imaging Radar-C (SIR-C) experiment, in April 1994. L (24 cm)- and C (5 cm)-band SAR data at HH (horizontal transmitting and receiving), HV (horizontal transmitting, vertical receiving), and VV (vertical transmitting and receiving) polarizations both in ground range and slant range and in two different passes were analyzed. The kappa statistic was used to identify meaningful texture measures to discriminate seven classes. The results show that the classifications of land use based only on tonal averages produced a kappa coefficient only slightly higher than 0.50. A kappa threshold of 0.90 was reached with the simultaneous inclusion of 15 texture measures for the six images (two bands, three polarizations). It was also found that the inclusion of texture features when only one band and one polarization was used could produce kappa values higher than 0.85. ©Elsevier Science Inc. 1997

INTRODUCTION

Discrimination among agricultural targets for improving crop forecasting is one of the most important applications of remote sensing from space (McDonald and Hall, 1980). This is especially true for large developing countries like Brazil, where local statistics lack cost-effective-

ness. Optical data from Landsat TM series has proven to be valuable for crop discrimination (Epiphanio, 1988), but frequent cloud cover during the growing season makes it difficult to acquire useful data on an operational basis. The use of high spatial resolution SAR data is expected to help overcome this problem due to the all-weather capability of radar systems.

Because of the coherent nature of echo recording, SAR images contain speckle noise, which represents very fine undesirable texture. Unfortunately, speckle increases the uncertainty of both spatial and radiometric interpretation. Visual analysis of SAR images suggest that the average backscatter of individual agricultural fields does not always allow discrimination among crops. In some cases, depending on the mix of crops, time of year, and feasibility of multitemporal coverage, the tonal averages might allow agricultural land use discrimination. One method to improve the discrimination capability of SAR images is the use of speckle filters based on local statistics (Lopes et al., 1990). However, speckle reduction results in some information loss such as texture and also problems arise for fields with a small number of pixels. The speckle in the SAR signal is partially removed by using multilook averaging of the data. The standard high resolution SIR-C data is 4-look averaged. Since the process of multilooking can also average out the texture features in the image, by multilooking, we also partially remove the texture in the image. Speckle is assumed to be a multiplicative factor to the texture signature in the SAR signal. Therefore, the absolute values of the texture features carry some contribution of the speckle, and it is difficult to separate the two completely. Ulaby et al. (1986) considered speckle to be independent of the textural variations associated with the spatial relationships of the scattering properties of distributed targets. Then it is

* INPE—Instituto Nacional de Pesquisas Espaciais, São José dos Campos, Brazil

Address correspondence to João Vianeí Soares, INPE—Instituto Nacional de Pesquisas Espaciais, Caixa Postal 515, 12201-970 São José dos Campos, SP, Brazil.

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assumed that the intrinsic scene variability (inhomogeneities of the terrain) causing the natural texture in the image can be separated even in the presence of speckle.

Texture is an important feature used in visual interpretation and therefore texture descriptors increase the performance of digital classification algorithms. The contribution of texture to the discrimination of crops and different types of forests has been demonstrated in previous work (see, e.g., Ulaby et al., 1986; Anys and He, 1995). Yanasse et al. (1993), attempting to separate forest from deforested areas using C-band SAR data, showed that tonal averages of homogeneous fields were inadequate, but discrimination was possible when the coefficient of variation was used. For small fields, the idea of extracting the texture features for previously segmented blocks, such that the edge effects are avoided for pixels in a vicinity of a field border, seems worthy of exploration.

Texture is not easy to quantify since there is neither a consensus on its definition nor a precise mathematical formulation for it. While the spectral information describes the gray level variation of a pixel, the texture deals with the spatial distribution of the gray levels over a segment of an image (Ulaby et al., 1986; Marceau et al., 1989). Nevertheless, various authors have tried to quantify texture. Methods for texture analysis have been developed using frequency patterns (Chen, 1990), first-order statistics (Hsu, 1978; Irons and Petersen, 1981), and second-order statistics (Haralick et al., 1973; Welch et al., 1990). Ulaby et al. (1986), using 16- and 24-look L-Band Seasat images, selected two second-order texture features to accomplish a 90% classification accuracy among five types of forest. Briand and Gray (1989), using SIR-B imagery, showed that first-order texture features can be used to improve the identification of land use classes. They stated that when only the tonal information was used, the total accuracy was 60% for their case study, while increasing to 85% by adding texture features. Texture measurements do not necessarily make assumptions about the statistical distribution of the data. If no assumptions are made, texture measurements can be taken from either radar or optical imagery with no restrictions.

With this in mind, methods for selecting texture measures to maximize the discrimination of agricultural land use classes in SAR images are investigated in this study. The data used in this investigation were taken in two overpasses during the April 1994 flight of the SIR-C/X-SAR campaign. In order to verify the advantage of using multiple frequencies and polarizations, the images selected were a combination of L- and C-bands, HH, HV, and VV polarizations (and incidence angles), processed in both ground range (GR) and slant range (SR) projections. The images were recorded in 256 gray levels. One of the GR products is a resampling of the SR representation, such that using them as different sets is

an attempt to verify, if any, the effect of the resampling on the texture features.

TEXTURE MEASUREMENTS

Texture measures based on the gray-level cooccurrence matrices (GLCM) are among the most used (Haralick et al., 1973). Each element $P(i,j)_{\Delta x, \Delta y}$ of the gray-level cooccurrence matrix represents the relative frequency with which two neighboring pixels separated by a distance of Δx columns and Δy lines occur, one with gray tone i and the other with gray tone j . Such matrices of gray-level spatial-dependence frequencies are a function of the angular relationship between the neighboring resolution cells as well as function of the distance between them. The main drawback of many texture methods, and of the GLCM approach in particular, is the large computer resources required. For example, for an 8-bit image (256 gray level), the cooccurrence matrix has 65,536 elements. To reduce this problem, one can use the texture measures extracted from the gray-level difference vector (GLDV) and from the sum and difference histogram (SADH) (Weszka et al., 1976; Unser, 1986; Welch et al., 1990). The GLDV approach is based on the absolute differences between pairs of gray levels i and j at a distance Δx columns and Δy lines apart at angle ϕ with a fixed direction. The SADH approach replaces the second-order statistics probability function of a cooccurrence matrix with estimates of the first-order probability functions along the principal axes of the cooccurrence matrix (Unser, 1986; Welch et al., 1990). Each element of the sum vector is defined as

$$P^S(k)_{\Delta x, \Delta y} = \sum_i \sum_j P(i,j)_{\Delta x, \Delta y} \quad \forall i+j=k$$

while each element of the difference vector is defined as

$$P^D(l)_{\Delta x, \Delta y} = \sum_i \sum_j P(i,j)_{\Delta x, \Delta y} \quad \forall |i-j|=l.$$

Using these techniques on the same 8-bit image produces a sum vector with 511 elements and a difference vector with 256 elements, considerably less than that needed by the GLCM approach.

The SADH method has the following advantages:

1. Nine of the GLCM textural features can be computed directly from the SADH approach.
2. The five other second-order features can be approximated by assuming mutual independence between $P^S(k)_{\Delta x, \Delta y}$ and $P^D(l)_{\Delta x, \Delta y}$ (Welch et al., 1990).

Many of the 14 textural features that can be extracted from the GLCM (representing different aspects of the matrix) are correlated (Ulaby et al., 1986). The texture features extracted from the GLCM that are investigated in this research are presented below. The physical meaning of some of the most relevant features is also given. For a full theoretical description of those GLCM texture

features, the reader is referred to Baraldi and Parmiggiani (1995):

a. Energy:

$$ENE_{\Delta x, \Delta y} = \sum_i \sum_j P(i, j)_{\Delta x, \Delta y}^2.$$

Energy is also called angular second moment (Haralick et al., 1973). Energy measures textural uniformity, that is, pixel pairs repetition; when the image patch under consideration is homogeneous (only similar gray level pixels are present), energy reaches its maximum, equal to 1. Thus, high values of energy occur when the gray level distribution over the block (or window) has either a constant or a periodic form. Energy is strongly uncorrelated to first-order statistical variables such as contrast and variance, such that energy may reach its maximum either with maximum or no variance and contrast values.

b. Entropy:

$$ENT_{\Delta x, \Delta y} = - \sum_i \sum_j P(i, j)_{\Delta x, \Delta y} \log P(i, j)_{\Delta x, \Delta y}.$$

Entropy measures the disorder of an image. When the image is not texturally uniform, entropy is very large. Conceptually energy and entropy are inversely correlated.

c. Contrast:

$$CON_{\Delta x, \Delta y} = \sum_i \sum_j (i - j)^2 P(i, j)_{\Delta x, \Delta y}.$$

Spatial frequency is the difference between the highest and the lowest values of a contiguous set of pixels. The GLCM contrast is the spatial frequency when the displacement vector is one. A low contrast patch presents a low value of the GLCM contrast. First-order statistics contrast and GLCM contrast are strongly related.

d. Local homogeneity:

$$HOM_{\Delta x, \Delta y} = \sum_i \sum_j \frac{P(i, j)_{\Delta x, \Delta y}}{1 + (i - j)^2}.$$

Local homogeneity is called inverse difference moment in Baraldi and Parmiggiani (1995). In fact, this parameter measures image homogeneity as it assumes larger values for smaller gray tone differences in pair elements. Then, it is easy to understand that GLCM contrast and local homogeneity are strongly, but inversely, correlated. However, this is not true in any case; local homogeneity decreases if contrast increases while energy is kept constant. On the other hand, local homogeneity decreases if energy increases while contrast remains constant.

e. Correlation:

$$COR_{\Delta x, \Delta y} = \frac{\sum_i \sum_j ij P(i, j)_{\Delta x, \Delta y} - \mu_i \mu_c}{\sigma_i \sigma_c}.$$

Correlation is a measure of gray-tone linear-dependencies in the image. High correlation values imply a linear relationship between the gray levels of pixel pairs. Thus, the GLCM correlation is uncorrelated to GLCM energy and entropy, that is, to pixel pairs repetitions. Correlation reaches its maximum regardless of pixel pair occurrence, as high correlation can be measured either in low or in high energy situations.

f. Chi-square:

$$CHI_{\Delta x, \Delta y} = \sum_i \sum_j \frac{P(i, j)_{\Delta x, \Delta y}^2}{P(i, \cdot)_{\Delta x, \Delta y} P(\cdot, j)_{\Delta x, \Delta y}},$$

where

$$\mu_i = \sum_j i P(i, \cdot)_{\Delta x, \Delta y}, \quad \mu_c = \sum_j j P(\cdot, j)_{\Delta x, \Delta y},$$

$$\sigma_i = \sqrt{\sum_j (i - \mu_i)^2 P(i, \cdot)_{\Delta x, \Delta y}},$$

$$\sigma_c = \sqrt{\sum_j (j - \mu_c)^2 P(\cdot, j)_{\Delta x, \Delta y}},$$

$$P(i, \cdot)_{\Delta x, \Delta y} = \sum_j P(i, j)_{\Delta x, \Delta y}, \quad P(\cdot, j)_{\Delta x, \Delta y} = \sum_i P(i, j)_{\Delta x, \Delta y}.$$

Chi-square can be understood as a normalization of the scene energy to the gray-tone linear dependencies in the image. It is then correlated to energy.

The following features were extracted using the sum and difference vectors technique:

g. Mean of the sum vector:

$$MS_{\Delta x, \Delta y} = \sum_k k P^S(k)_{\Delta x, \Delta y}.$$

h. Variance of the sum vector:

$$VS_{\Delta x, \Delta y} = \sum_k (k - MS_{\Delta x, \Delta y})^2 P^S(k)_{\Delta x, \Delta y}.$$

i. Energy of the sum vector:

$$ENES_{\Delta x, \Delta y} = \sum_k P^S(k)_{\Delta x, \Delta y}^2.$$

j. Entropy of the sum vector:

$$ENTS_{\Delta x, \Delta y} = \sum_k P^S(k)_{\Delta x, \Delta y} \log P^S(k)_{\Delta x, \Delta y}.$$

l. Mean of the difference vector:

$$MD_{\Delta x, \Delta y} = \sum_l l P^D(l)_{\Delta x, \Delta y}.$$

m. Variance of the difference vector:

$$VD_{\Delta x, \Delta y} = \sum_l (l - MD_{\Delta x, \Delta y})^2 P^D(l)_{\Delta x, \Delta y}.$$

n. Energy of the difference vector:

$$ENED_{\Delta x, \Delta y} = \sum_l P^D(l)_{\Delta x, \Delta y}^2.$$

o. Entropy of the difference vector:

$$ENTD_{\Delta x, \Delta y} = - \sum_l P^D(l)_{\Delta x, \Delta y} \log P^D(l)_{\Delta x, \Delta y}.$$

Regardless of the method and texture features extracted

(GLCM or sum and difference vector approach), entropy is a measure of disorder in a scene (randomness in a sub-region). It is largest for equally distributed gray levels and smallest when they are unequally distributed. Energy is also a measure of local homogeneity. Local homogeneity and contrast are inverse measures of each other. The correlation also expresses local homogeneity. Welch et al. (1990), attempting to separate cloudy regions from mountain regions using these texture features, found that these regions were well discriminated.

Measures other than those derived from the gray-level cooccurrence matrices are also useful for texture-based discrimination. Soares et al. (1987) achieved good characterization of the spatial behavior of agricultural fields from C-band airborne scatterometer using the autocorrelation function, a second-order feature. Here, we used the spatial autocorrelation with lags (0,1), (1,0), and (1,1). The autocorrelation with lag (1,1) refers to the correlation calculated between each pixel and its neighbor to the right and below ($\varphi = -45^\circ$). The spatial autocorrelation function at a lag l_i and l_j is

$$AUT_{l_i l_j} = \frac{\sum_{i=1}^{n_i-l_i} \sum_{j=1}^{n_j-l_j} (x_{i,j} - \bar{x})(x_{i+l_i, j+l_j} - \bar{x})}{\sum_{i=1}^{n_i} \sum_{j=1}^{n_j} (x_{i,j} - \bar{x})^2},$$

where l_i and l_j are the lags in columns and lines respectively, n_i and n_j are the number of lines and columns, x_{ij} is the gray level of the pixel located at the i th line and j th column, and $\bar{x}$ is the average gray level.

Other parameters used to quantify texture of images are based on first-order statistics, that is, they are calculated on the subregion, not within a specific or moving window. The features describing distribution of data based on the first-order histogram are extensively well known. In this work, the following first-order statistics were calculated: variance, coefficient of variation, asymmetry, asymmetry of the mean to the median, and kurtosis, defined as:

Variance:

$$V = \sum_i [i - M]^2 P(i),$$

where $P(i)$ is the relative frequency of a gray level i and M is the mean obtained by

$$M = \sum_i i P(i).$$

Coefficient of variation:

$$CV = \frac{\sqrt{V}}{M}.$$

Asymmetry:

$$AS = \frac{\sum_i |(i - M)^3| P(i)}{V^{3/2}},$$

Asymmetry of the mean to the median MED of the gray levels:

$$ASM = \frac{|M - MED|}{\sqrt{V}}$$

Kurtosis:

$$CUR = \frac{\sum_i (i - M)^4 P(i)}{V^2}$$

Yanasse et al. (1993), analyzing the statistical properties of C-band data acquired over the Amazon region, tested several distributions including K -multilook magnitude, gamma, square root of gamma, and log-normal, to determine those that best fit the data. They found that the parameter a of the K -multilook magnitude (amplitude) distribution expressed texture differences between forest and nonforest classes well. High values of a indicate homogeneity within a region. Two estimators for a were also included in this investigation; one based on K -multilook intensity distribution and the other based on the K -multilook magnitude distribution (Yanasse et al., 1993).

Assuming a multiplicative model, SAR images can be seen as a product of the natural texture of the terrain and the fine grain texture dictated by the speckle noise. Adjacent pixels are spatially correlated in terms of speckle, such that the autocorrelation function using neighboring pixels is expected to be masked as a descriptor of the natural texture of a target.

DATA COLLECTION AND ALGORITHM DEVELOPMENT

The study area is the Bebedouro SIR-C/X-SAR Supersite (Jet Propulsion Laboratory, 1993) located on the left bank ($9^\circ 07'S$, $40^\circ 18'W$ Gr) of the São Francisco River, a major north-south trending flowing river system. The river crosses a vast semiarid region in northeast Brazil, where a government sponsored development program will irrigate over 1 million hectares for agriculture, of which 200,000 have already been irrigated. In the Bebedouro Irrigation Project (BIP), individual farms range from 5 ha to 12 ha and the total irrigated area is 1750 ha (Soares et al., 1988). Crops and orchards in the BIP include mango, vine, tomato, melon, and watermelon. Other types of land cover (found at the time of the SIR-C April 1994 flight) are pasture, pasture with bushes, and bare soil.

The analysis was done on data taken in two overpasses during the April 1994 flight. The main characteristics of the data used are summarized in Table 1. Images processed in both ground range GR (GR1 and GR2) and slant range SR projections were used. Initially the SR image were multilook processed in order to reduce speckle noise and to obtain approximately square pixels ($12.57 \text{ m} \times 13.32 \text{ m}$). Each set is composed of six images

Table 1. Characteristics of the SAR Images Used as Obtained by the SIR-C Instrument

	<i>GR1</i>	<i>GR2</i>	<i>SR</i>
Acquisition date	04/09/94	04/13/94	04/13/94
Band	L and C	L and C	L and C
Polarization	HH, HV, VV	HH, HV, VV	HH, HV, VV
Projection	Ground range	Ground range	Slant range
Incidence angle	47.77°	37.97°	37.98°
Resolution	12.5 m×12.5 m	12.5 m×12.5 m	4.19 m×13.32 m

(combination of L- and C-bands, HH, HV, and VV polarizations). The images were recorded in 256 gray levels. The GR2 and SR came from the same raw data, the GR being a resampling of the SR representation. Using them as different sets is an attempt to determine whether the resampling is efficient for delineating the texture features.

Field work simultaneous to the overpasses produced a land cover map to be used as ground truth. Based on acreage, distribution, and importance of each crop, seven classes were defined: 1) vine, 2) melon/watermelon, 3) tomato, 4) bare soil, 5) pasture, 6) pasture with bushes, and 7) mango orchard. Melon and watermelon were grouped as a single class because of their similarities in terms of soil cover, density of seeding, and plant height. Well-kept pasture and pasture with abundant shrubs were considered as different classes. A total of 225 plots have been considered in this investigation: 7 mango orchards, 10 watermelon/melon, 22 pasture, 48 pasture with bushes, 34 bare soil, 13 tomato and 91 vine. An agronomic characterization was done for every sampled plot following the procedure established by Cihlar et al. (1987). Vineyards were planted with a 3.5 m spacing among individual plants, and plant height was approximately 1.7 m with all possible orientations for different fields; situations of leaves off as part of the agricultural practice were also found. Melon and watermelon individual plants were 2 m×1.5 m in area, they covered from 50% to 80% of the soil, soil roughness was low between plants. Tomato was planted with 0.8 m between lines and 0.5 m between individuals, soil roughness was high, and the soil was being irrigated. Bare soil represented plots with smooth roughness conditions being prepared for cropping. Pasture was well-kept grasses completely covering the soil used for cattle ranching. Pasture with bushes accounted for pasture in bad conditions of management with a great amount of invasive plants (bushes and weeds). Mango orchards were planted on a 10 m×10 m spacing and included ages from young (2 m high) to mature stands (5–8 m high); for some of the young ones melon and watermelon plantations were found between individual trees. As one can depict from the above description of the agricultural cover themes, within classes variability was also present in the fields. Then the inhomogeneities of the terrain caused by different plant

canopies, density of trees and seeding, orientation, agronomic stage, plant height, and soil conditions both in terms of wetness and roughness are responsible for the natural texture in SAR images, to be accounted for in order to increase the discrimination capability of SAR images.

For each property, a local map was created using a topographic chart of the BIP as a cartographic base. The property scale maps were joined, and the resulting regional land cover map was digitized and registered to the SAR images. Typically remote sensing images are registered to the base map, but the reverse was done to avoid distortion in the gray levels, preserving the image in its original format.

A number of 25 measures were calculated in each of the images composing a set: 14 second-order measures extracted from the gray-level cooccurrence matrices and the sum and differences vectors, three spatial autocorrelations, five based on first order statistics, two estimators for α , and the mean, totaling 150 measures per set (GR1, GR2, and SR). Table 2 summarizes the measures.

One can evaluate the performance of a subset of texture features through its effectiveness in classifying targets. An "optimal" subset can be defined as the one that assures 100% of accuracy. Measures of divergence (distances) or similarity are widely employed in the classification process (Richards, 1986). The algorithms based upon Euclidean distance are the most commonly used because they do not require that the data have a specific distribution. When the data are normally distributed the Mahalanobis distance can be used instead of the Euclidean distance, because it relies not only on the mean but also on the covariance matrix.

If the true class to which the target belongs is known, one can build up a confusion matrix relating assigned class to ground truth. Usually the classification process is evaluated by the fraction of samples correctly classified, which is represented by the ratio between the sum of the elements in the diagonal of the confusion matrix and the total number of classified samples. One alternative method of evaluation is the kappa coefficient of agreement (Rosenfield and Fitzpatrick-Lins, 1986; Hudson and Ramm, 1987; Foody, 1992), which has the advantage of considering all the elements in the confusion matrix and not only the diagonal ones. The kappa coefficient

Table 2. Analyzed Variables (Mean and 24 Texture Measures)

Abbreviation	Full Name of the Variable
CON	Contrast
COR	Correlation
CHI	Chi-square
ENT	Entropy
HOM	Local homogeneity
ENE	Energy
MS	Mean of sum vector
VS	Variance of the sum vector
ENES	Energy of the sum vector
ENTS	Entropy of the sum vector
MD	Mean of the difference vector
VD	Variance of the difference vector
ENED	Energy of the difference vector
ENTD	Entropy of the difference vector
AUT _{0,1}	Spatial autocorrelation with lag (0,1)
AUT _{1,0}	Spatial autocorrelation with lag (1,0)
AUT _{1,1}	Spatial autocorrelation with lag (1,1)
V	Variance
CV	Coefficient of variation
AS	Assymetry
ASM	Assymetry of the mean to the median
CUR	Kurtosis
Alfam	α estimator to K multilook magnitude
Alfai	α estimator to K multilook intensity
M	Mean

cient attempts to estimate the proportion of accurate classification by assuming some random misclassification of the targets. The maximum value of kappa coefficient is 1 when all targets are accurately classified. Negative values are possible and low values of kappa indicate that the classification algorithm did not obtain good results (Landis and Koch, 1977).

Using 150 measures for characterizing a target is neither practical nor necessary because of the tremendous computing time needed to extract them (Rennó, 1996) and the correlation existing between several of them (Ulaby et al., 1986; Baraldi and Parmiggiani, 1995); a well-chosen subset should produce equal results. Two methods for selecting texture variables are used in this study: 1) using an optimization technique that selects variables from the original set of 150; 2) using a preselection stage based on the variances among classes that diminishes the new working set to about 35% of the original one. The description of the two methods follows.

Method 1

As described above each set is composed of six images (two bands, three polarization). For each image, 25 measures (texture features and the mean) were computed, resulting in 150 variables for every set. As the variables have different absolute ranges, they were normalized to have a mean of zero and unitary variance.

Initially the discriminatory capability of each variable was computed using the kappa coefficient. The variable

with the highest individual kappa was the first retained in the selection procedure. Then, a second variable was selected among the remaining 149, choosing the one that, when combined with the first variable, gave the highest kappa, in a step by step operation. The process went on until a kappa of at least 0.90 was obtained for the first time. Since it is assumed that there is a strong correlation between many variables derived from the GLCM (Ulaby et al., 1986; Baraldi and Parmiggiani, 1995), after selecting a new variable at a given step, combination among the already selected was done to verify if any of them could be eliminated without dropping the kappa. This procedure is analogous to that of stepwise variable selection in multivariate regression analysis.

Method 2

Despite the fact that the methodology concerning Method 1 appears to be robust (many combinations are tested), there is no guarantee that the best subset is selected since all possible combinations of the 150 variables are not tested. Variables that vary greatly between land use classes are more efficient at discriminating between those classes, and hence should have high kappa coefficients. The variance of a variable among classes is defined as

$$VAR_{class} = \frac{\sum_i n_i (\bar{x}_i - \bar{x})^2}{N},$$

where $\bar{x}_i$ is the mean of the variable for the class i , n_i is the number of samples of that class, $\bar{x}$ is the overall mean of the variable, and N is the total number of samples. As all variables were reduced to have mean of zero, the variance among classes becomes:

$$VAR_{class} = \frac{\sum_i n_i \bar{x}_i^2}{N}.$$

Using the variance among classes, a preselection was done to reduce the number of variables for the final selection. In a first step, the variables were sorted according to their variance among classes. Later, for each selected variable, the samples were classified and the kappa coefficient obtained. The preselection was stopped when the kappa coefficient reached the value of 1. Then the selection of variables was done as described in Method 1, but with the considerable advantage of reducing the computation and analysis efforts by as much as 65%.

Since the image sets GR1, GR2, and SR were not coregistered, the analysis and selection of texture features were done separately. The procedure described above for both Method 1 and Method 2 is valid for all the three sets.

Although some variables did not have Gaussian distribution, the Mahalanobis distance was adopted in the

classification process, because preliminary tests indicated that the results using this measure outperformed the Euclidean distance (Rennó, 1996). This is probably due to the presence of strong correlations among variables, which are dismissed by the Euclidean distance.

The classification was done using all 225 samples divided in seven classes, as described above. Unfortunately, it was not possible to use different training and test samples, because some of the crop classes were represented by a small amount of samples over the test site (seven for mango orchard, 10 for watermelon, 13 for tomato). This might introduce an overestimation of the overall classification performance, since the confusion matrices are calculated using training samples.

In order to verify the existence of any implicit relationship among the selected measures of each set, we also attempted to classify all the sets using the variables chosen for the other sets independently. This was done since it was expected that the amounts of information delivered by different groups of variables are comparable.

Although the analysis done in this investigation takes advantage of the unique capability of the SIR-C/X-SAR instruments to provide multifrequency and multipolarization SAR imagery, one has to keep in mind that in the practical world, multifrequency multipolarization data is rarely available. Operational systems such as RADARSAT, JERS-1, and ERS-2 provide only a single band and polarization. For this reason, in this investigation, we also tested the usefulness of the texture measures for each image individually.

Although some features of the GLCM are correlated (Rennó, 1996; Ulaby et al., 1986), as well as some first-order to some second-order statistics (Baraldi and Parmigiani, 1995), in this investigation no *a priori* choice of texture features were made, since the correlation among them were calculated at each step in the selection procedure, such that one feature would not be included in the set if the combination with one previously retained (strongly correlated to it), did not add much to the accuracy of the discrimination. Baraldi and Parmigiani (1995) state that similar results are expected for energy and entropy clustering, since these two variables are strongly, although inversely correlated, but energy is to be chosen over entropy because it has a normalized range. In this investigation, as stated earlier, all the features were normalized to have a mean of zero and unitary variance.

Once the true classes are informed before the computational routine starts, such that the kappa values can be calculated at each step, the process of selecting features is fully automated. The computational platform used in this investigation was the "interactive data language (IDL)" and listings can be obtained from the authors.

RESULTS AND DISCUSSION

As a remainder, we will refer to the selection process using all 150 variables (25 measures times six images for each set, Table 1) as Method 1; the procedure using a preselection based on the variance among classes, VAR_{class} , will be referred as Method 2. The use of Method 2 allowed the number of variables to be reduced to about 35% of the total (64 for GR1, 55 for GR2, and 54 for SR). The criteria used in the preselection stage was to obtain a kappa coefficient of agreement of 1 regarding the classification of all samples.

Figure 1 presents a comparison of the classification performance using the mean (the average gray level for each sample) and the texture features selected by Methods 1 and 2. The performance is given by the kappa coefficient of agreement corresponding to the subset of variables added up step by step. One can notice that the means show a performance comparable to a subset of features (texture and one or more means), when only a comparable number of variables were selected. Also, it can be seen that the use of six means (two bands, three polarization) increases the discriminatory power of radar data, showing that multifrequency and multipolarization are important in agricultural land cover discrimination. Yet, as each set is composed by six images, one can use only six means resulting, for the study area, in a maximum value of 0.5 for the kappa coefficient of agreement, which indicates only a moderate accuracy of the classification method (Landis and Koch, 1977). When texture features are added, one can achieve kappa values superior to 0.90 by taking 15 measures simultaneously. One must keep in mind that a threshold of 0.90 for the kappa value was established in the step by step selection procedure. Also, the texture features were always obtained for previously segmented blocks such that edge effects are avoided. This most probably helped making it possible to achieve high values of kappa. As stated in the previous section, there might be an overestimation of that performance because the overall classification were done using training samples.

Methods 1 and 2 showed very similar performances. It is important to say that the variables selected by the two methods are not necessarily the same. According to this result one could choose Method 2 over Method 1 because it is three times faster. From now on, for simplicity, the discussion is directed to the results obtained by Method 2 only.

Figure 2 shows the relationship between the variance among classes and the kappa coefficient of agreement for each individual variable (measure or texture feature). The vertical lines indicate the minimum values of the variance VAR_{class} of the preselected variables, such that the accumulated kappa coefficients become 1. The variables to the left of the minimal VAR_{class} vertical lines are rejected in the preselection stage. The

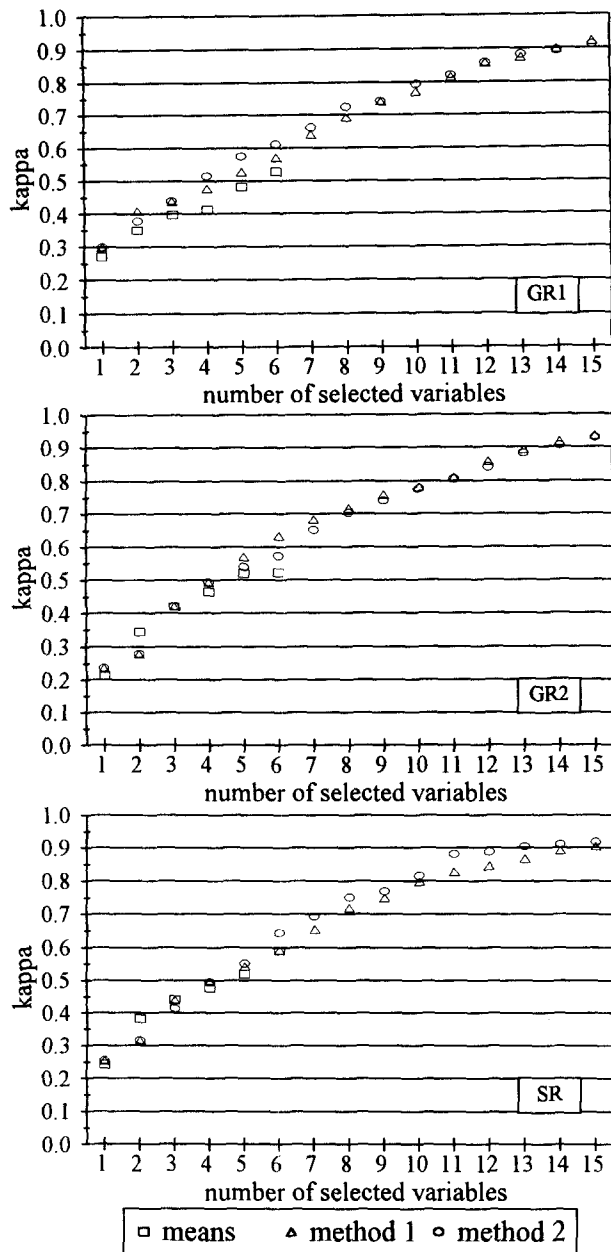


Figure 1. Performance of the classification procedure measured by the kappa coefficient of agreement as a function of the number of variables using the mean and texture features selected by Methods 1 and 2.

reader should notice that a great amount of variables are discarded in this stage. One can also remark that the highest values of kappa and VAR_{class} come from the L-band, showing that by adding texture measures to the mean tone the discriminatory potential is increased in this frequency. On the other hand, when only the selected variables are shown (Fig. 3), one observes that the best set of selected variables does not correspond to the variables with highest individual kappa; that is the reason for not rejecting too many variables in the pre-selection phase.

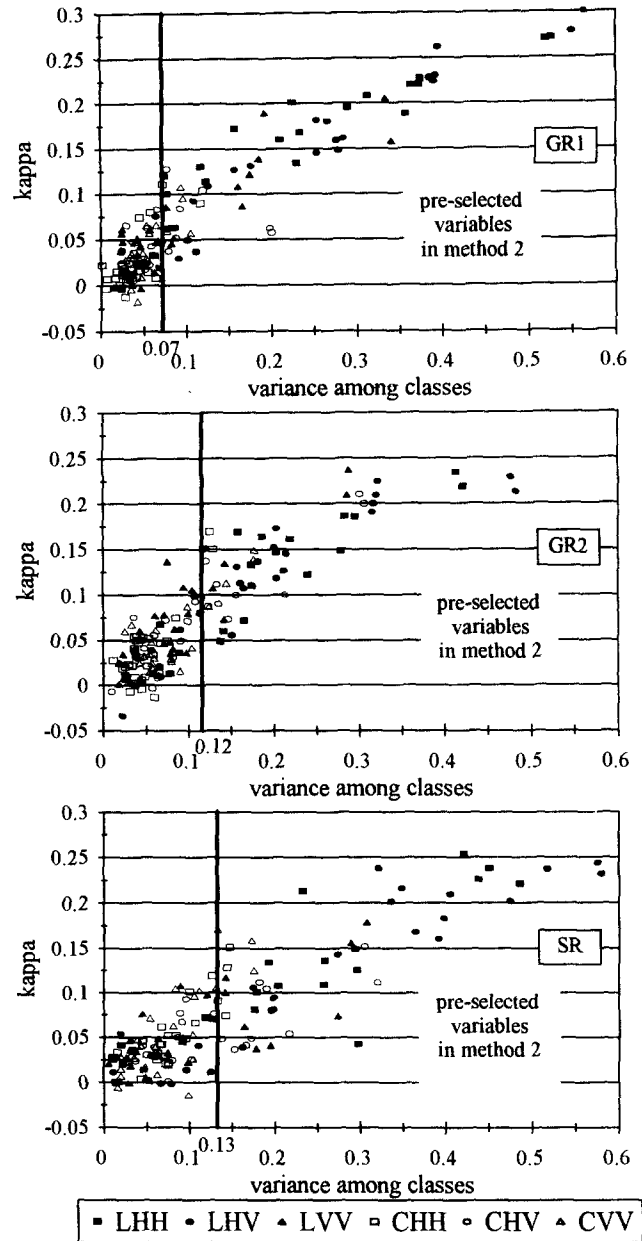


Figure 2. Relationship between the variance among classes and the kappa coefficient of agreement for each individual variable.

Each image appeared with at least one texture feature selected except for CVV (C-band, VV polarization) for data set GR2. The L-band images were, in general, the ones that most contributed with more variables retained. Regarding the set GR1, the VV polarization played an important role coming up with a substantial amount of texture information, while this was not the case for the GR2 and SR sets.

Table 3 shows the list of selected variables for all three sets of images, adopting as criterion a kappa coefficient of agreement of 0.90. Again, it is noticeable that the tonal mean was an important discriminatory measure since most of the images are represented by the mean.

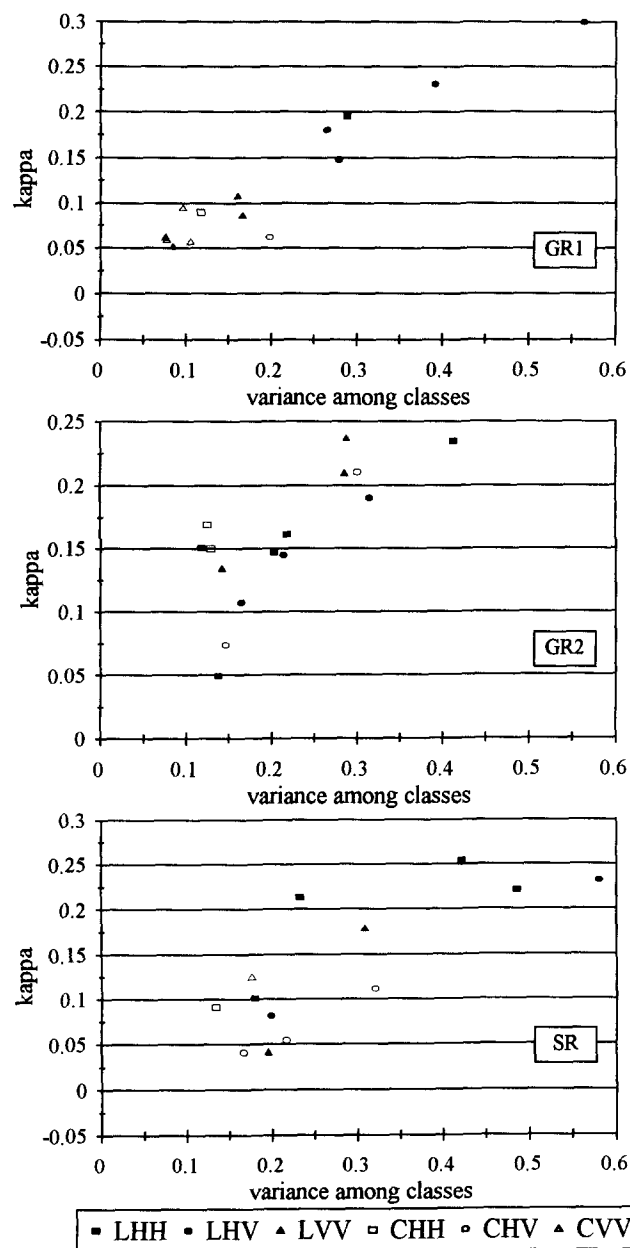


Figure 3. Relationship between the kappa coefficient of agreement and the variance among classes for each individual measure selected, using Method 2.

Each set (GR1, GR2, SR) has a distinct subset of selected texture features, but, nevertheless, some of them are highly correlated. For example, the following selected variables have an intrinsic coefficient of correlation higher than 0.90: *mean and mean of the sum vector*, *entropy of the difference vector and mean of the difference vector*, *energy of the difference vector and chi-square*, *contrast and mean*, *variance of sum vector and variance*.

The confusion matrix concerning the group GR2 is shown in Table 4. It shows that out of 225 samples of the seven major classes of land cover on the Bebedouro Irrigation Project under analysis in this work, only 11 of

Table 3. Selected Measures (Mean and Texture Features) Using Method 2

GR1	GR2	SR
CHLHH	CORLHH	CONLHH
ENTSLHV	CHLHH	HOMLHH
ENTDLHV	MLHH	MSLHH
MLHV	ASMLHH	CURLHH
VLHV	CURLHH	MSLHV
CONLVV	CONLHV	ASLHV
CHILVV	VSLHV	MSLVV
HOMLVV	MDLHV	VDLVV
CVLVV	MSLVV	VDCHH
MCHH	ENEDLVV	ENEDCHV
MSCHV	MLVV	MCHV
MSCVV	MSCHH	CVCHV
ENEDCVV	MCHH	MSCVV
ENTDCVV	MSCHV	
CVCVV	ASMCHV	

The last three letters refer to the contributing band and polarization.

them were misclassified. The corresponding map (obtained from the classification) is shown in Figure 5, the ground truth map is shown in Figure 4. Figure 6 shows the misclassified samples with respect to the true classes. All of them are overlaid on the GR2 LVH image. The agreement is also reflected by a kappa coefficient close to 0.90.

The selection procedure seems to be adequate to extract meaningful texture features to help discriminating among agricultural land cover themes, such as those present on the study site. In addition, the optimization process can be considered as an efficient one, since out of the 150 variables, one could bound the analysis on about 35% of them and only 15 were retained when the threshold of 0.90 of kappa value was reached. A next step in this analysis is to build a supervised classifier that uses the texture features added to the tonal means over a priori segmented blocks.

Although Table 3 indicates that we cannot generalize about a set of texture measurements ideal for agricultural land cover discrimination, we have attempted, in order to verify the existence of any implicit relationship among the selected measures of each set, to classify all of them using the chosen variables of the others sets separately. The results of this classification are given on Table 5. The kappa coefficients were higher than or equal to 0.75, showing that, in general, the subset of variables (texture features or texture measures) retained for one individual dataset (GR1, GR2, and SR) can be used to classify another independent group, even though with lesser accuracy. According to Landis and Kock (1977), kappa values between 0.60 and 0.80 indicate good accuracy in the classification process.

As to how the texture relates to the structural properties of crops, there is hardly anyway to find out. Texture by definition carries information about the spatial relationships between pixels within a certain window

Table 4. Confusion Matrix of the Classification Obtained for the GR2 Group

	Mango	Water-melon	Pasture	Pasture with Bushes	Bare Soil	Tomato	Vine
Mango	7	0	0	0	0	0	0
Watermelon/melon	0	10	0	0	0	0	0
Pasture	0	0	21	0	0	0	1
Pasture with bushes	0	0	0	45	0	0	3
Bare soil	0	0	1	1	31	0	0
Tomato	0	0	0	0	0	13	0
Vine	0	0	0	2	3	0	87

Lines refer to the classification and columns to the real cover themes.

while backscatter values of each pixel carries information about the structure of the crop canopy. In order to explore the physical significance of the textural features, the set chosen for GR2 is investigated here: Out of the 15 measures retained three of them represented tonal means (LHH, LVV, and CHH), four represented second-order features that are correlated with tone (mean of the difference vector LHV, mean of the sum vector LVV, mean of the sum vector CHH and mean of the sum vector CHV), others are three first-order features (Kurtosis LHH, asymmetry of the mean to the median LHH and asymmetry of the mean to the median CHV), and the remaining five are others second-order features (correlation LHH, chi-square LHH, contrast LHV, energy of the difference vector LVV, variance of the sum vector LHV). It becomes clear that the tonal averages (that imply the need for multifrequency and multipolarization) are important since they appeared three times and also because they are correlated to the means of the sum and difference vectors that were present four times. The theoretical description of correlation, chi-square, contrast, energy, and variance, the five other textural measures selected was given in a previous section. Certainly, both the scattering properties and the inhomogeneities of the terrain were accounted for with the combination of tone, first- and second-order texture features. Another noticeable result is that L-band contributed more than C-band in the studied case, appearing with 11 out of the 15 features selected for both the two bands. In Table 6, we present examples of means of texture features for three classes: vineyards, bare soil and pasture. *Contrast* is highest for vineyards, lowest for pasture; *en-*

trophy is also highest for vineyards but lowest for bare soil, *homogeneity* and *energy* are both lowest for vineyards and highest for pasture, and *correlation* is lowest for vineyards and highest for pasture. One should always remember that texture features were normalized to have mean 0; that explains the negative numbers. Although it is not easy to associate structural properties of the canopies with spatial patterns vineyards presented the highest texture (highest contrast, lowest correlation, highest entropy, lowest homogeneity and energy). This reflects the expectation that double bounce scattering from vineyards, as stakes are used to hold the individual plants, combined with observed differences in leafy cover due to different stages of plant development, modulate both tone and second-order relationships. This is also in agreement with the expectation that spatial variations concerning well-kept pasture completely covering the soil and smooth bare soil to be dominated basically by speckle (less entropy and contrast, more correlation, homogeneity and energy). Pasture presented the lowest texture, although close to bare soil ones. The pasture fields were homogeneous, and the bare soil fields varied from smooth to rough and from moist to dry within a particular field. There is quite a range in canopy structure and spatial patterns that contribute to a combination of backscattering properties (regarding different frequencies and polarizations), and texture features such as energy, contrast, homogeneity, and correlation. Ulaby et al. (1986), combining tone with second-order contrast and local homogeneity for L-band HH SIR-A imagery, achieved classification accuracies higher than 90% for five types of forest. They did not attempt to explain how the scattering properties of different types of forest focused in their case relate to the spatial relationships of the pixels. In the analysis of their results, Baraldi and Parmiggiani (1995) related only four image surface themes with texture types: 1) very high texture (mountains); 2) high/medium texture (sea ice with floes); 3) low texture (clouds); and 4) no texture (continental ice). The physical interpretation in that case seems to be easier.

So far we have discussed only the addition of textural features to both bands and polarizations information. However, in the practical world, multifrequency

Table 5. Comparison of the Classification Performance of the Three Sets Using Their Own Selected Measures (Main Diagonal) to the Ones Using the Other Two Sets, Independently

	Kappa		
	GR1	GR2	SR
Variables from GR1	0.91	0.81	0.80
Variables from GR2	0.82	0.91	0.85
Variables from SR	0.75	0.82	0.90



■ bare soil ■ pasture ■ tomato ■ water melon
 ■ mango ■ vine ■ pasture with bushes

Figure 4. Ground truth land use map of the Bebedouro Irrigation Project, overlaid on the LHV image of set GR2.



■ bare soil ■ pasture ■ tomato ■ water melon
 ■ mango ■ vine ■ pasture with bushes

Figure 5. Land use map of the Bebedouro Irrigation project from the sample classification using texture features added to the mean, overlaid on the LHV image of set GR2.

multipolarization data is rarely available. We then performed the analysis of each image individually. One must recognize that only 25 measures were available (24 texture features and the mean). Figure 7 displays the performance of the Mahalanobis distance classifier for each image as a function of the number of variables selected, always measured by the kappa coefficient of agreement. This noteworthy result shows that even by using only one single frequency and polarization, the land use classes of agricultural areas can be discriminated using the texture features, regardless of the frequency and polarization used, with kappa values as high as 0.85. This result must be interpreted with care, since it seems to convey the idea that all the channels are equivalent and to contradict the statement made above that multifrequency and multipolarization are important for crop classification. The reason that the optimizer reached the same range of kappa values for all polarizations and frequencies, only implies the fact that inhomogeneity of the terrain under study and the spatial relationships of pixels within each block are the most significant feature in discriminating the classes. This does not anticipate that for other land covers such as different mix of crops or forests, multifrequency, and polarization data are not required. In fact, Figure 1 shows that the individual SIR-C channels give low kappa values (around 0.2), and by adding the tonal averages for all channels, one can reach a kappa value higher than 0.5, showing that multifrequency and multipolarization are important. Keeping in mind that the classification were done over segmented blocks and no

independent test samples were used is also important to understand the real meaning of the results.

Although the approach used in our investigation is original and different from previous work, our results are in agreement with previous results on the role of texture

Figure 6. Misclassified samples with respect to the true classes, overlaid on the LHV image of set GR2.



■ Misclassified samples

Table 6. Examples of Second Order Texture Features for Vineyards, Bare Soil, and Pasture

	Second-Order Texture Features				
	Contrast	Correlation	Entropy	Homogeneity	Energy
Vineyards	0.66	-0.48	0.21	-0.72	-0.24
Bare soil	-0.25	0.28	-0.24	0.35	0.19
Pasture	-0.59	0.38	-0.17	0.68	0.21

The features were normalized to have a mean of zero and unitary variances.

features in crop and forest discrimination in SAR images. Anys and He (1995) showed that the integration of texture measures with three multipolarization channels improved the rate of correct classification by 9.79% for the crops. Ulaby et al. (1986), using both first- and second-order spatial statistics to enhance the classification of forest categories, showed that second-order statistics outperformed the results to tonal and first-order statistics on Seasat radar images. For SIR-A imagery, the combined use of tone and texture resulted in overall classification accuracies exceeding 90%. Briand and Gray (1989), using SIR-B imagery, showed that first-order texture features can be used to improve the identification of land use classes. They found that the total accuracy arised from 60% to 85% when texture features were added to the tone mean. Yanasse et al. (1993), attempting to separate forest from deforested areas using C-band SAR data, showed that tonal averages of homogeneous fields were inadequate, but discrimination was possible when the coefficient of variation was used.

CONCLUSIONS

The results of this investigation confirmed the ones published in previous work that the addition of texture measures improves the accuracy of classification of agricultural cover themes from SAR imagery. For the study case, it was possible to reach a kappa threshold of 0.90, established in the optimization process of selecting texture features in a stepwise-like operation. In spite of that, one cannot make generalizations about a set of texture measurements as being the ideal one for the discrimination of agricultural targets. It has been demonstrated that the different characteristics of the images tested in terms of projection and incidence angle has not influenced much in the selection of the texture features, since a subset selected for one data take or resampling from ground to slant range projection, could be applied to the others independently with good results.

It became evident that the tonal information was very important in the discrimination of cover themes, since, for most of the cases analyzed, the means or other variables highly correlated to them were selected for all images. A related conclusion is that multifrequency and multipolarization do improve the classification performance by providing independent images, as expected

from the physics of matter-microwaves interaction. When only the tonal means were considered, the combination of all of them allowed the overall classification performance, given by the kappa values, to increase from about 0.20 (negligible accuracy) for individual channels to close to 0.60 (moderate accuracy) for the combination of all channels together.

Another noteworthy result came out when the analysis of images at one band and polarization was carried out individually. Although the tonal information exhibits poor discriminatory capability when used alone, the addition of texture features made it possible to achieve high values of coefficient of agreement, indicating that classifiers built up using texture features along with the tone are expected to render the discrimination of agricultural cover classes using single channel SAR systems, such as RADARSAT and JERS-1, a feasible task. On the other hand, this result does not mean that all channels convey similar information. It also has to be analyzed in light of the conclusion related to the integration of all tone averages described in the previous paragraph. In fact, this result only implies the fact that inhomogeneity of the terrain under study and the spatial relationships of pixels within each block are the most significant features in the discrimination of the classes analyzed. Again, to extend this result to other areas with different mix of crops or other cover types such as forests, specific studies have to be carried out.

Even though one must keep in mind that there might be an overestimation of the kappa coefficients of agreement, since, as stated above, confusion matrices were calculated using training samples, texture proved to be a key discriminator of agriculture crops using SAR imagery. It is important to remember that since the main objective of this work was the selection of feature vectors for discriminating classes, and no independent set of samples are used for verifying the classifier, the high values of kappa only imply the best choice of feature for the area under study.

It has been demonstrated that the different characteristics of the images tested in terms of projection and incidence angle has not influenced much in the selection of the texture features, since a subset selected for one data take or resampling from ground to slant range projection, could be applied to the others independently with good results.

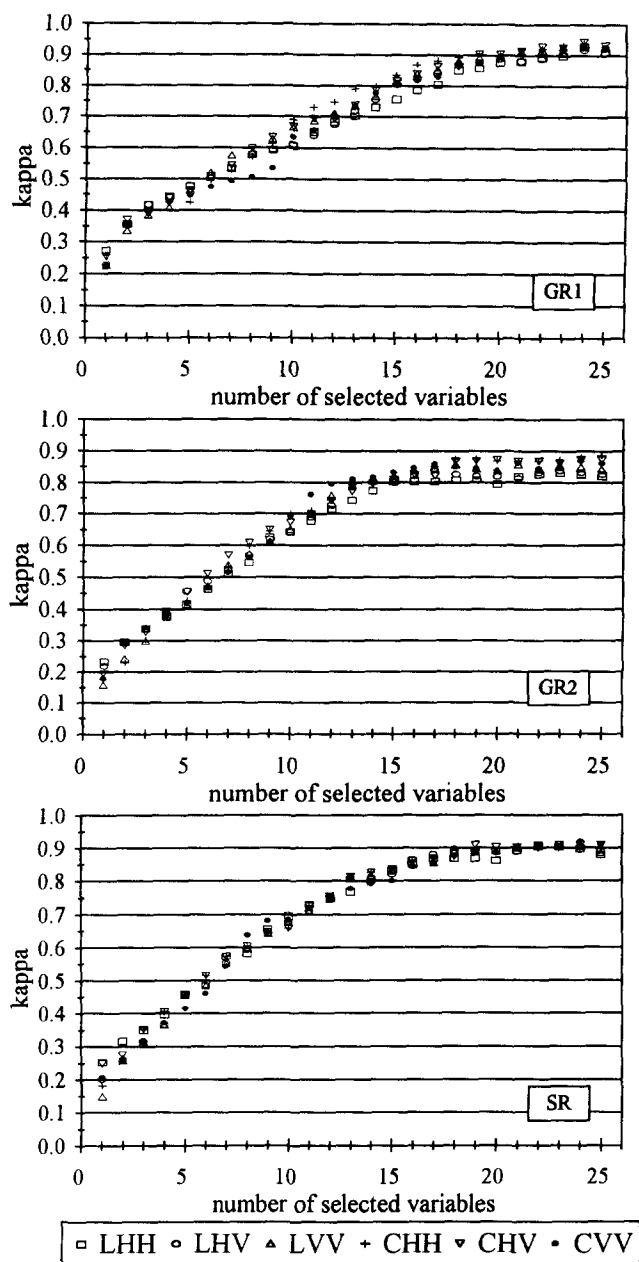


Figure 7. Performance of the classification methodology, considering the six images (two bands, three polarizations) separately.

It is very difficult to compare the results found in this research with previous work, because of the very specific goal to come out with a procedure for optimizing the selection of texture features. Previous research using texture features to increase land cover discrimination, such as the ones by Ulaby et al. (1986) and Anys and He (1995), did not attempt to select the best texture features to describe cover classes but to evaluate the value of adding texture features to tonal classifiers. Ulaby et al. (1986) achieved classification accuracies higher than 90% for five types of forest when second-order spatial statistics were combined with tone information. We avoided

comparison of the results of this investigation with pixel by pixel classifiers since the classification itself was a crucial part of the optimization procedure, not the ultimate goal. Moreover, pixel by pixel classification algorithms designed to account for the *a priori* probability of classes, which is another way of considering the information from the neighbors, do not take the spatial relationships among pixels into consideration. Again, speckle fading is a major problem in tone based classification.

The methodology used for selecting the texture features based on the preselection of variables, using its variance among land use classes as a guideline, was shown to be very efficient, since the accuracy of the discrimination was high and the number of variables to be accounted for was considerably smaller. To the best of our knowledge, this is an interesting contribution of this study to the literature on the subject. This procedure could be used to build a supervised classifier based on *a priori* segmented images such that training blocks, representing the classes on the image, are used to extract the texture features vectors using the optimization technique developed here. Then one could obtain the measures of divergence (distance) of texture features for the classification process. This method can be applied, in some cases, when land cover maps are available at scales that allow registration with SAR images, from which segments are promptly obtained. At the moment, segmentation algorithms for SAR imagery is one of our major efforts of investigation.

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